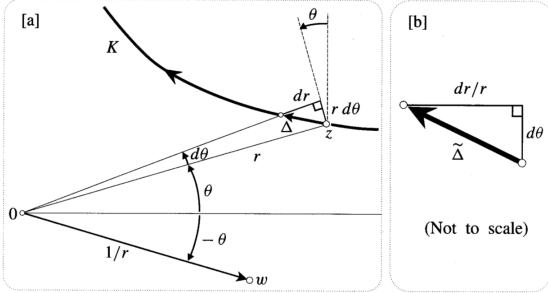


Chapter 4

From Winding Numbers to Topological Degrees

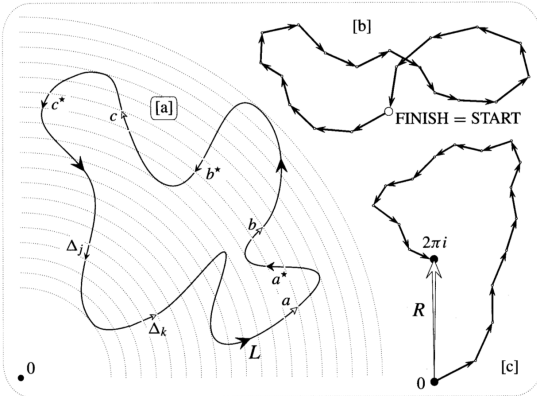
Example 4.1 (Geometry of the line integral of complex inversion: arbitrary closed curves). Consider $\int_{\beta} \frac{1}{z} dz$ where β is an arbitrary closed curve.



From the plot above, we have

$$\Delta = r d\theta e^{i(\theta + \frac{\pi}{2})} + dr e^{i(\theta + \pi)}, \quad w = \frac{1}{r} e^{-i\theta},$$

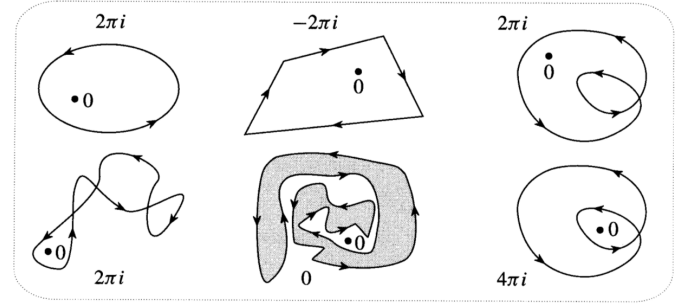
where (r, θ) are the polar coordinates of a point on the curve. Therefore we have $w\Delta = -\frac{dr}{r} + i d\theta$. As shown in subplot [a] below, for a simple closed curve β , each Δ_j can be paired up with another Δ_k such that the values $\frac{dr}{r}$ of $w_j\Delta_j$ and $w_k\Delta_k$ have the same magnitude but the different sign. Therefore, the real part of the integral $\int_{\beta} \frac{1}{z} dz$ vanishes.



The starting point and the ending point of a closed curve have the same value of r . In contrast, it is a different story for θ . If β does not enclose the origin, then the starting point and the ending point have the same value of θ and the contour integral reduces to 0, as shown in subplot [b].

However, if β does enclose the origin, we have $\int_{\beta} \frac{1}{z} dz = 2\pi i$, with 2π being the difference of the polar coordinate θ for the ending and starting points.

Finally, for an integral on an arbitrary closed curve, the Seifert decomposition (Lemma 3.68) implies that the integral can be expressed as the sum of integrals on a number of simple closed curves, as illustrated below.



4.1 Winding numbers

Definition 4.2. The *winding number* of an oriented closed curve $\Gamma \subset \mathbb{C}$ around a point $z \in \mathbb{C} \setminus \Gamma$ is

$$w(\Gamma, z) := \frac{1}{2\pi i} \int_{\Gamma} \frac{d\gamma}{\gamma - z}, \quad (4.1)$$

where γ is a parametrization of Γ .

Lemma 4.3. Any winding number in (4.1) is an integer.

Proof. Define $\phi(t) := \exp\left(\int_0^t \frac{\gamma'(s)}{\gamma(s) - z} ds\right)$ where $t \in [0, 1]$ and Γ is parameterized as $\gamma : [0, 1] \rightarrow \mathbb{C}$. We have

$$\begin{aligned} \phi'(t) &= \frac{\gamma'(t)\phi(t)}{\gamma(t) - z} \Rightarrow \left[\frac{\phi(t)}{\gamma(t) - z} \right]' = \frac{\phi'(\gamma - z) - \phi\gamma'}{(\gamma - z)^2} = 0 \\ \Rightarrow \frac{\phi(1)}{\gamma(1) - z} &= \frac{\phi(0)}{\gamma(0) - z} \Rightarrow \phi(1) = \frac{\phi(0)(\gamma(1) - z)}{\gamma(0) - z} = 1, \end{aligned}$$

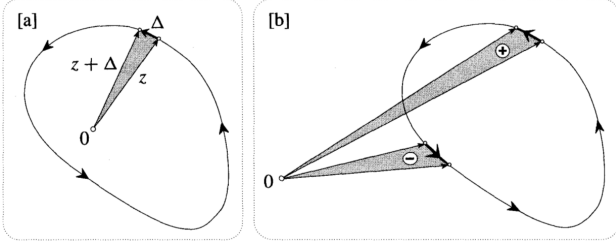
where the last equality follows from $\phi(0) = 1$ and the closeness $\gamma(0) = \gamma(1)$. The rest of the proof follows from

$$\phi(1) = \exp(w(\Gamma, z)2\pi i) = 1 \implies w(\Gamma, z) \in \mathbb{Z}. \quad \square$$

Lemma 4.4. The winding number (4.1) is a constant in each connected component of $\mathbb{C} \setminus \Gamma$. In particular, we have $w(\Gamma, z) = 0$ as $z \rightarrow \infty$, i.e., the constant is 0 in the unbounded component.

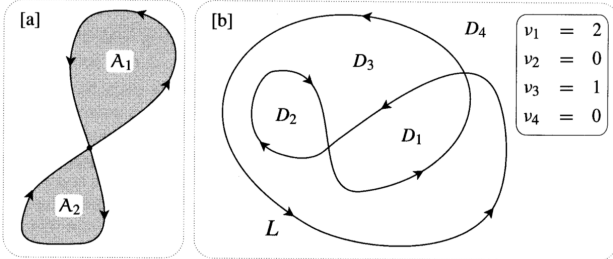
Proof. $w(\Gamma, z)$ is analytic with respect to z , and thus w is continuous in terms of z . Any two points in a component of $\mathbb{C} \setminus \Gamma$ can be connected by a path inside the component, hence Lemma 4.3 dictates that $w(\Gamma, z)$ be a constant for that component. As $z \rightarrow \infty$, the modulus of the integral in (4.1) can be made arbitrary small and thus by Lemma 4.3 it must be zero. $\square$

Example 4.5 (Geometry of the line integral of complex conjugate). Consider $\int_{\alpha} \bar{z} dz$ for a simple closed curve α .



Since $\frac{1}{2} \text{Im}(\bar{z}\Delta)$ is the area of the triangle shown in subplot [a], $\frac{1}{2} \text{Im}(\int_{\alpha} \bar{z} dz)$ equals the signed area of the simple closed region inside α . Furthermore, $\text{Arg}(\bar{z}) = \text{Arg}(z^{-1})$. By arguments similar to those in Example 4.1, we deduce that $\text{Re}(\int_{\alpha} \bar{z} dz) = 0$. These arguments also hold when the origin is outside of α , as shown in subplot [b]. Therefore we conclude that $\int_{\alpha} \bar{z} dz$ equals $2i$ times the signed area of the simple closed region bounded by α .

This result easily generalizes to arbitrary closed curves. As shown in subplot [a] below, it is possible that $\int_{\alpha} \bar{z} dz$ vanishes if $A_1 = -A_2$.



In more general cases such as that in subplot [b], we have

$$\int_{\alpha} \bar{z} dz = 2i \sum_j \nu_j A_j, \quad (4.2)$$

where A_j is the signed area of the j th complement D_j of α and ν_j is the winding number of L around a point in D_j , cf. Lemma 4.4.

Lemma 4.6. The winding number of a positively oriented circle Γ with its center at a and its radius as r is

$$w(\Gamma, z) = \begin{cases} 1 & \text{if } |z - a| < r; \\ 0 & \text{if } |z - a| > r. \end{cases} \quad (4.3)$$

Proof. Parametrize Γ as $\gamma(t) = a + re^{it}$ with $t \in [0, 2\pi]$. By (4.1), we calculate $w(\Gamma, a) = 1$ and deduce the first clause from Lemma 4.4, which also implies the second clause. $\square$

Exercise 4.7. Denote $\mathbb{S}^1 = \{e^{it} : t \in [0, 2\pi]\}$ and consider a closed curve Γ as the image of $\mathbb{S}^1$ under the function

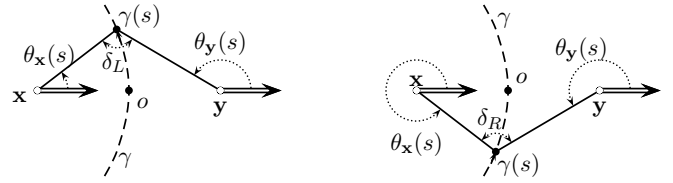
$$\gamma(t) = z + r(t)e^{i\theta(t)}, \quad (4.4)$$

where θ is the angle of $\gamma(t)$ in the polar coordinate system whose pole is at z . Show that $w(\Gamma, z) = \frac{1}{2\pi}(\theta(2\pi) - \theta(0))$. According to this result, what is the meaning of the winding number $w(\Gamma, z)$?

Definition 4.8 (Alexander [1928]). Let γ be a closed curve and $p(t)$ a particle moving continuously in the plane. Suppose p crosses γ at time t_c . A point $\gamma(s_c) = p(t_c)$ is called a *positive crossing* if, in a sufficiently small open neighborhood of s_c , γ is oriented from the left of the directed pathline of p to its right, and a *negative crossing* otherwise.

Lemma 4.9. Let Γ be a closed curve parametrized and oriented by $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$. Then a positive/negative crossing of a moving particle p increases/decreases $w(\Gamma, p(t))$ by 1, respectively.

Proof. We only prove the case of a negative crossing since the other case is symmetric.



Referring to the above figure, let $o = p(t_0) \in \gamma$ be the crossing point and denote $\mathbf{x} = p(t_0 - \tau)$, $\mathbf{y} = p(t_0 + \tau)$. As $\tau \rightarrow 0$, the direction of the ray $\mathbf{x} \rightarrow \mathbf{y}$ converges to that of $\mathbf{u}(o, t_0)$, which splits the plane into two halfplanes: $\theta_{\mathbf{y}}(s) = \theta_{\mathbf{x}}(s) + \delta_L$ in the left one and $\theta_{\mathbf{y}}(s) = \theta_{\mathbf{x}}(s) - \delta_R$ in the right one. Choose the basepoint of γ as $o = \gamma(0)$, $\gamma(2\pi) = o$, then $\lim_{s \rightarrow 0} \delta_L = \lim_{s \rightarrow 2\pi} \delta_R = \pi$ for a single traversal of γ . Finally, Exercise 4.7 yields

$$\begin{aligned} w_{\gamma}(\mathbf{y}) - w_{\gamma}(\mathbf{x}) &= \frac{1}{2\pi}(\theta_{\mathbf{y}}(2\pi) - \theta_{\mathbf{x}}(2\pi) + \theta_{\mathbf{x}}(0) - \theta_{\mathbf{y}}(0)) \\ &= \frac{1}{2\pi}(-\pi - \pi) = -1. \end{aligned} \quad \square$$

Lemma 4.10. If a particle crosses an oriented closed curve a finite number of times as it moves continuously from $\mathbf{x}$ to infinity, the winding number $w_{\gamma}(\mathbf{x})$ is equal to the number of negative crossings minus the number of positive crossings.

Exercise 4.11. Prove Lemma 4.10 by induction.

Theorem 4.12. Suppose the cycle decomposition of an oriented closed curve Γ as in Lemma 3.68 consists of a finite number of Jordan curves. Then for each $\mathbf{x} \notin \Gamma$, its winding number $w(\Gamma, \mathbf{x})$ is the number of counterclockwise Jordan curves containing $\mathbf{x}$ minus that of clockwise Jordan curves containing $\mathbf{x}$.

Exercise 4.13. Prove Theorem 4.12.

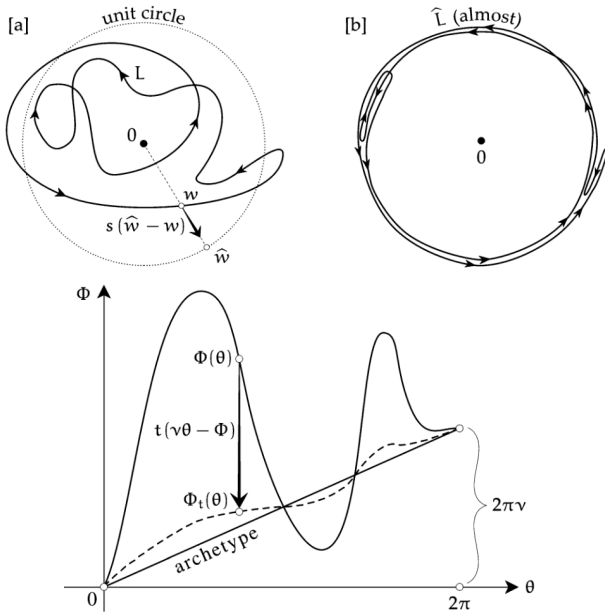
4.2 The Hopf degree theorem

Lemma 4.14. Any closed curve Γ that satisfies $0 \notin \Gamma$ and $w(\Gamma, 0) = \nu \neq 0$ can be continuously deformed into the archetypal loop $\hat{\gamma}_\nu = e^{i\nu\varphi}$ with $\varphi \in [0, 2\pi]$, and vice versa.

Proof. Parametrize Γ as $\gamma(\theta) = r(\theta)e^{i\Phi(\theta)}$ with $\theta \in [0, 2\pi]$ and define a normalized version of Γ as $\tilde{\gamma}(\theta) = e^{i\Phi(\theta)}$. Construct a free homotopy

$$H(s, t) := \begin{cases} (1 - 2t)\gamma(2\pi s) + 2t\tilde{\gamma}(2\pi s) & \text{if } t \in [0, \frac{1}{2}]; \\ \exp\{i\Phi_t(\theta)\} & \text{if } t \in [\frac{1}{2}, 1], \end{cases}$$

where $\Phi_t(\theta) := (2 - 2t)\Phi(2\pi s) + (2t - 1)2s\nu\pi$. Then Γ is continuously deformed to $\tilde{\gamma}([0, 2\pi])$ and then to $\hat{\gamma}_\nu([0, 2\pi])$, as shown below.



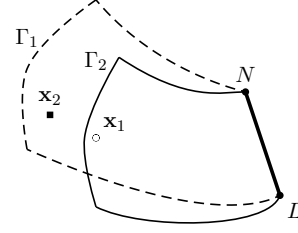
The free homotopy from the archetypal loop to Γ can be constructed similarly. $\square$

Exercise 4.15. For the proof of Lemma 4.14, give another free homotopy in which the modulus and the argument changes simultaneously from 0 to 1.

Theorem 4.16 (Hopf [1935]). For any given point $\mathbf{x} \in \mathbb{C}$, two closed curves Γ_1 and Γ_2 are freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}\}$ if and only if $w(\Gamma_1, \mathbf{x}) = w(\Gamma_2, \mathbf{x})$.

Proof. The sufficiency follows directly from Lemma 4.14 while the necessity from Lemma 4.4 and the free homotopy being a continuous function. $\square$

Example 4.17. Two loops Γ_1 (dashed lines) and Γ_2 (thin solid lines) share as their common arc $\overline{LN}$ (the thick line segment). In $\mathbb{C}$, they are of the same homotopy class $H(s, t) = t\Gamma_1(s) + (1 - t)\Gamma_2(s)$ with the fixed basepoint as $L = \Gamma_1(0) = \Gamma_1(1) = \Gamma_2(0) = \Gamma_2(1)$; they are also freely homotopic in $\mathbb{C}$.



Furthermore, they are freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}_1\}$ because of the Hopf theorem 4.16 and $w(\Gamma_1, \mathbf{x}_1) = w(\Gamma_2, \mathbf{x}_1)$. However, they are not freely homotopic in $\mathbb{C} \setminus \{\mathbf{x}_2\}$ because $w(\Gamma_1, \mathbf{x}_2) \neq w(\Gamma_2, \mathbf{x}_2)$. In other words, there is no way Γ_1 can be continuously deformed to Γ_2 in $\mathbb{C} \setminus \{\mathbf{x}_2\}$, or, any continuous deformation of Γ_1 to Γ_2 in $\mathbb{C}$ must cross $\mathbf{x}_2$.

4.3 Characteristic points of analytic functions

4.3.1 Roots or Zeros

Definition 4.18. A subset $\mathcal{M} \subseteq \mathcal{X}$ in a metric space $(\mathcal{X}, d)$ is *discrete* iff

$$\forall x \in \mathcal{X}, \exists \epsilon > 0 \text{ s.t. } \forall y \in \mathcal{M} \setminus \{x\}, d(x, y) > \epsilon. \quad (4.5)$$

Such a set consists of *isolated points*.

Lemma 4.19. A subset $\mathcal{M} \subseteq \mathcal{X}$ is *discrete* in $\mathcal{X}$ if and only if there is no limit point of $\mathcal{M}$ in $\mathcal{X}$.

Proof. By Definition 2.16, the existence of a limit point x of $\mathcal{M}$ in $\mathcal{X}$ implies

$$\exists x \in \mathcal{X} \text{ s.t. } \forall \epsilon > 0, \exists y \in \mathcal{M} \text{ s.t. } d(x, y) \in (0, \epsilon).$$

The negation of the above statement yields (4.5). $\square$

Definition 4.20. A point $a \in \mathcal{D}$ is called a *zero* or a *root* of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ iff $f(a) = 0$. A zero point a of f is of *multiplicity* m iff $\exists m \in \mathbb{N}^+$ such that

$$f^{(m)}(a) \neq 0 \text{ and } \forall j = 0, 1, \dots, m-1, f^{(j)}(a) = 0.$$

In particular, a zero of multiplicity 1 is a *simple zero* or a *simple root*.

Example 4.21. The zeros of $f(z) = \sin z - 1$ form the set $\{\frac{\pi}{2} + 2k\pi : k \in \mathbb{Z}\}$. Each zero is of multiplicity 2.

Lemma 4.22. Let Ω be a domain and $a \in \Omega$ a zero point of an analytic function $f : \Omega \rightarrow \mathbb{C}$. The following two statements are equivalent:

- (a) The multiplicity of a is m .
- (b) $f(z) = (z - a)^m \varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$.

Proof. This follows directly from Theorem 3.112. $\square$

Theorem 4.23. Let Ω be a domain. If an analytic function $f : \Omega \rightarrow \mathbb{C}$ is not identically zero, then the set of its zeros is discrete in Ω .

Proof. Let a be a zero of f with multiplicity m . Lemma 4.22 yields

$$f(z) = (z - a)^m \varphi(z),$$

where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$. By the continuity of φ , there exists some $R > 0$ such that for all $z \in \mathcal{U}_R(a)$ we have $\varphi(z) \neq 0$. The proof is completed by Definition 4.18. $\square$

Example 4.24. The zeros of $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0; \\ 0, & x = 0, \end{cases} \quad (4.6)$$

are $0, \pm \frac{1}{n\pi}$ where $n \in \mathbb{N}^+$ and 0 is a limit point of the zeros. Hence Theorem 4.23 does not hold for real functions.

4.3.2 Critical points

Definition 4.25. A *critical point* of a continuous function of a single real variable is a value in the domain of f at which either f is not differentiable or the derivative of f vanish.

Example 4.26. A regular value of a curve $\gamma : (0, 1) \rightarrow \mathbb{R}^n$ is a regular point of the curve, cf. Definition C.379. For a parameter $t \in [0, 1]$ to be a critical point of γ , the tangent vector must be either zero or undefined.

Definition 4.27. A *critical point* of a differentiable function $f : \Omega \rightarrow \mathbb{R}^n$ where $\Omega \subset \mathbb{R}^m$ is a point in Ω where the rank of the Jacobian matrix of f is not maximal. A point $p \in \mathbb{R}^n$ is a *regular value* of f if none of its preimages is critical; otherwise it is a *critical value* of f .

Theorem 4.28 (Sard). For a C^k function $f : \Omega \rightarrow \mathbb{R}^n$ with $\Omega \subseteq \mathbb{R}^m$ and $k \geq \max(m - n + 1, 1)$, the set of critical values of f has Lebesgue measure 0 in $\mathbb{R}^n$.

Lemma 4.29. For an analytic function $f : \Omega \rightarrow \mathbb{C}$ with $\Omega \subseteq \mathbb{C}$ open, its Jacobian determinant $\det f'(z_0)$ is positive at any z_0 that is not a critical point of f .

Proof. The Jacobian of $f(x + iy) = u(x, y) + iv(x, y)$ is

$$f' = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}.$$

The rest follows from the Cauchy-Riemann theorem 2.50 and Definition 4.27. $\square$

Corollary 4.30. Any critical point $a \in \Omega$ of an analytic function $f : \Omega \rightarrow \mathbb{C}$ satisfies $f'(a) = 0$.

Proof. By the proof of Lemma 4.29, the only possibility for f to satisfy $\det f'(a) = 0$ is to have $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0$ at a . Hence the conclusion holds. $\square$

Example 4.31. The continuous function $h(x + iy) = x - iy^3$ satisfies $\det J_h = -3y^2$. All points on the real axis are critical points as a result of crushing in the vertical direction.

4.4 The argument principle

Notation 3. $\text{BCJ}(\Gamma)$ and $\text{UCJ}(\Gamma)$ denote the bounded and unbounded complements of a Jordan curve Γ , respectively.

Corollary 4.32. A positively oriented Jordan curve Γ has

$$w(\Gamma, p) = \begin{cases} 1 & \text{if } p \in \text{BCJ}(\Gamma); \\ 0 & \text{if } p \in \text{UCJ}(\Gamma). \end{cases} \quad (4.7)$$

Proof. This follows from Theorem 4.16 and Lemma 4.6. $\square$

Corollary 4.33. A negatively oriented Jordan curve Γ has

$$w(\Gamma, p) = \begin{cases} -1 & \text{if } p \in \text{BCJ}(\Gamma); \\ 0 & \text{if } p \in \text{UCJ}(\Gamma). \end{cases} \quad (4.8)$$

Proof. The winding number in Definition 4.2 depends on the orientation of Γ indicated by the parametrization γ . For the reciprocal curve Γ^- of Γ , we have $w(\Gamma^-, p) = -w(\Gamma, p)$. The rest follows from Corollary 4.32. $\square$

4.4.1 For polynomials

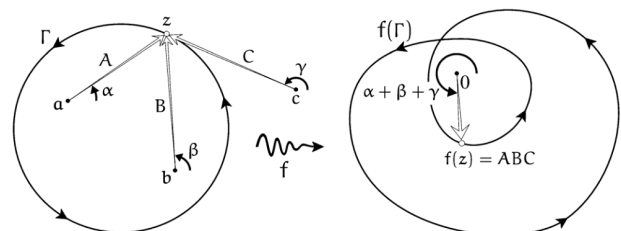
Lemma 4.34. Suppose a polynomial f has no roots on a positively oriented Jordan curve Γ . Then $w(f(\Gamma), 0)$ equals the number of roots of f in $\text{BCJ}(\Gamma)$, counted with their algebraic multiplicity.

Proof. Write $f(z) = \prod_{j=1}^{n_m} (z - z_j)^{m_j} \prod_{j=1}^{n_k} (z - y_j)^{k_j}$ where m_j and k_j are respectively the algebraic multiplicities of the roots z_j and y_j in $\text{BCJ}(\Gamma)$ and $\text{UCJ}(\Gamma)$. Definition 4.2 yields

$$\begin{aligned} w(f(\Gamma), 0) &= \frac{1}{2\pi i} \int_{f(\Gamma)} \frac{d\gamma}{\gamma} = \frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)}{f(z)} dz \\ &= \frac{1}{2\pi i} \int_{\Gamma} \left(\sum_{j=1}^m \frac{m_j}{z - z_j} + \sum_{j=1}^k \frac{k_j}{z - y_j} \right) dz \\ &= \sum_{j=1}^m m_j w(\Gamma, z_j) + \sum_{j=1}^k k_j w(\Gamma, y_j) \\ &= \sum_{j=1}^m m_j, \end{aligned}$$

where the last step follows from Corollary 4.32. $\square$

Example 4.35. For $f(z) = (z - a)(z - b)(z - c)$, we have $w(f(\Gamma), 0) = 2$, as shown below.



Suppose instead we had $f(z) = (z - a)^2(z - b)(z - c)$. Then this multiplicity would be taken care of by $f'(z)$ so that $w(f(\Gamma), 0) = 3$.

Exercise 4.36 (Alternative proof of the fundamental theorem of algebra). Prove Theorem 3.47 via the argument principle in Lemma 4.34.

4.4.2 For analytic functions

Lemma 4.37. Let $f, g : \Omega \rightarrow \mathbb{C}$ be analytic functions where $\Omega \subset \mathbb{C}$ is a simply connected region. Suppose $\Gamma \subset \Omega$ is an oriented closed curve such that $f(z)g(z) \neq 0$ for all $z \in \Gamma$. Then we have

$$w((fg)(\Gamma), 0) = w(f(\Gamma), 0) + w(g(\Gamma), 0). \quad (4.9)$$

Exercise 4.38. Prove Lemma 4.37.

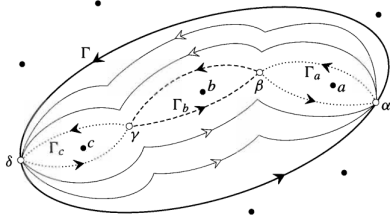
Theorem 4.39. Let Ω be a domain that contains a closed curve $\Gamma : [0, 1] \rightarrow \Omega$. For an analytic function $f : \Omega \rightarrow \mathbb{C}$ and a point $p \notin f(\Gamma)$, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} w(\Gamma, z_j) m_j, \quad (4.10)$$

where m_j is the algebraic multiplicity of the j th root z_j of the equation $f(x) = p$. In the particular case of Γ being a positively oriented Jordan curve, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} m_j. \quad (4.11)$$

Proof. Consider the case of Γ being a positively oriented Jordan curve. By the Hopf degree theorem 4.16 and the Seifert decomposition in Lemma 3.68, we can deform and decompose, without changing any $w(\Gamma, z_j)$, the Jordan curve Γ into a finite number of Jordan curves $\{\Gamma_j\}$ such that each Γ_j encircles at most one z_j and each z_j with $w(\Gamma, z_j) \neq 0$ is encircled by some Γ_j .



Then Definition 4.2 yields

$$(*) : \quad \begin{cases} w(\Gamma_j, z_j) &= w(\Gamma, z_j), \\ w(f(\Gamma), p) &= \sum_j w(f(\Gamma_j), p). \end{cases}$$

Since f is analytic on Ω , Theorem 3.112 dicates that, near each z_j ,

$$f(z) - p = K(z)g(z) \quad (4.12)$$

where $K(z) = \frac{f^{(m_j)}(z_j)}{m_j!} + \frac{f^{(m_j+1)}(z_j)}{(m_j+1)!}(z - z_j) + \dots$, $f^{(m_j)}(z_j) \neq 0$, and $g(z) = (z - z_j)^{m_j}$. It follows that

$$\begin{aligned} w(f(\Gamma_j), p) &= w(f(\Gamma_j) - p, 0) = w(K(\Gamma_j)g(\Gamma_j), 0) \\ &= w(K(\Gamma_j), 0) + w(g(\Gamma_j), 0) = w(g(\Gamma_j), 0) \\ &= m_j w(\Gamma_j, z_j) = m_j w(\Gamma, z_j) = m_j, \end{aligned}$$

where the first equality follows from Definition 4.2, the second from (4.12), the third from Lemma 4.37, the fourth from $f^{(m_j)}(z_j) \neq 0 \Rightarrow K(z_j) \neq 0$ and the Hopf theorem 4.16, the fifth from $g(z) = (z - z_j)^{m_j}$ and Lemma 4.34, the sixth from the first clause in (*), and the last from Corollary 4.32. Then the second clause in (*) implies (4.11).

Finally, (4.10) follows from (4.11) and Theorem 4.12. $\square$

Corollary 4.40. Let Ω be a domain and $\Gamma : [0, 1] \rightarrow \Omega$ be a positively oriented Jordan curve inside Ω . For analytic functions $f, h : \Omega \rightarrow \mathbb{C}$ and a point $p \notin f(\Gamma)$, we have

$$\frac{1}{2\pi i} \int_{\Gamma} h(z) \frac{f'(z)}{f(z) - p} dz = \sum_{z_j \in f^{-1}(p)} h(z_j) m_j, \quad (4.13)$$

where m_j is the multiplicity of the root z_j of $f(x) = p$.

Exercise 4.41. Prove Corollary 4.40.

Example 4.42. At a critical point z_c of an analytic function f , we have, from (4.12), $f(z) - f(z_c) = K(z)(z - z_c)^{m_c}$ where $m_c > 1$ and $K(z_c) \neq 0$. Hence f maps a positively oriented Jordan curve Γ around z_c to a closed curve $f(\Gamma)$ that winds m_c times around $f(z_c)$.

Lemma 4.43. For an analytic function $f : \Omega \rightarrow \mathbb{C}$ with $\Omega \subset \mathbb{C}$ open, the algebraic multiplicity of f at $a \in \Omega$ is 1 if and only if a is not a *critical point* of f .

Proof. Suppose $\det f'(z_0) \neq 0$. Then by the inverse function theorem f is a local homeomorphism: it maps an infinitesimal circle C_a centered at a to another infinitesimal circle centered at $p = f(a)$. Furthermore, Lemma 4.29 dictates that the two circles have the same orientation. Hence (4.10) yields $m_j(a) = 1$.

Suppose $\det f'(z_0) = 0$. Then f can be expressed in the form of (4.12), where, by Lemma 4.34, the algebraic multiplicity m_j of f at a does not equal 1. $\square$

Definition 4.44. The *signum function* or the *sign function* is the function $\text{sgn} : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$\text{sgn}(x) = \begin{cases} +1 & \text{if } x > 0; \\ 0 & \text{if } x = 0; \\ -1 & \text{if } x < 0. \end{cases} \quad (4.14)$$

Corollary 4.45. For a positively oriented Jordan curve Γ , an analytic map $f : \text{BCJ}(\Gamma) \rightarrow \mathbb{C}$, and a regular value p of f satisfying $p \notin f(\Gamma)$, we have

$$w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} \text{sgn} \det f'(z_j). \quad (4.15)$$

Proof. For each preimage z_j of a regular value p of f , Lemma 4.29 states that $\det f'(z_j) > 0$. The rest follows from (4.11) and Lemma 4.43. $\square$

Theorem 4.46 (Darboux–Picard). Suppose $f : \Omega \rightarrow \mathbb{C}$ is an analytic function where Ω is a domain that contains a Jordan curve $\Gamma : [0, 1] \rightarrow \Omega$. If the restriction $f|_{\Gamma}$ is injective, then so is $f|_{\text{BCJ}(\Gamma)}$.

Proof. Since Γ is Jordan and $f|_{\Gamma}$ is injective and continuous, $f(\Gamma)$ is also a Jordan curve. For any $p \in \text{BCJ}(f(\Gamma))$, (4.10) gives $w(f(\Gamma), p) = \sum_{z_j \in f^{-1}(p)} w(\Gamma, z_j) m_j$. By Lemma 4.4, Lemma 4.6, and the condition of Γ being a Jordan curve, $w(\Gamma, z_j) = 0$ for any $z_j \in \text{UCJ}(\Gamma)$ and $w(\Gamma, z_j) = 1$ for any $z_j \in \text{BCJ}(\Gamma)$. Consequently, $f^{-1}(p) \cap \text{BCJ}(\Gamma)$ must contain exactly one point; otherwise it would contradict (4.10). $\square$

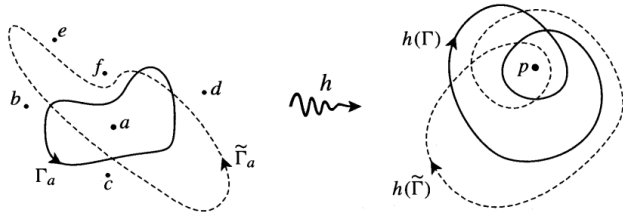
Exercise 4.47. Give an alternative proof of Theorem 4.46 by considering points in $\text{BCJ}(\Gamma)$.

4.4.3 For continuous functions

Definition 4.48. The *topological multiplicity* or the *local degree* or the *index* of a continuous map $h : \Omega \rightarrow \mathbb{C}$ at a point $a \in \Omega$ is

$$\nu_h(a) := w(h(\Gamma_a), p), \quad (4.16)$$

where Γ_a is a positively oriented Jordan curve Γ_a such that $a \in \text{BCJ}(\Gamma_a) \subseteq \Omega$ and $\text{BCJ}(\Gamma_a)$ contains no other preimages of $p = h(a)$.



Example 4.49. The topological multiplicity $\nu_h(a)$ of the continuous function $h(z) = \bar{z}$ is -1 for any $a \in \mathbb{C}$.

Exercise 4.50. Is it appropriate to illustrate Example 4.42 by the plot below Definition 4.48? If yes, explain why; otherwise state how to modify the plot to illustrates Example 4.42.

Definition 4.51. The *topological degree* of a continuous map $h : \text{BCJ}(\Gamma) \rightarrow \mathbb{C}$ at $p \in \mathbb{C} \setminus h(\Gamma)$ is given by

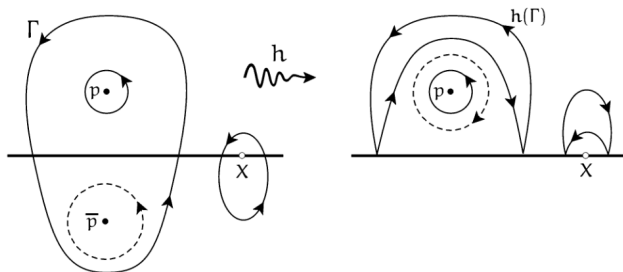
$$\deg(h, \text{BCJ}(\Gamma), p) := w(h(\Gamma), p). \quad (4.17)$$

Theorem 4.52. For a continuous map $h : \overline{\text{BCJ}(\Gamma)} \rightarrow \mathbb{C}$ and a point $p \in \mathbb{C} \setminus h(\Gamma)$, the topological degree of h at p equals the number of preimages of p in $\text{BCJ}(\Gamma)$, counted with their topological multiplicities,

$$\deg(h, \text{BCJ}(\Gamma), p) = \sum_{a \in h^{-1}(p)} \nu_h(a). \quad (4.18)$$

Proof. This follows from Definitions 4.48 and 4.51 and the first paragraph in the proof of Theorem 4.39. $\square$

Example 4.53. Depending on the choice of Γ , the topological degree of the continuous function $h(x + iy) = x + i|y|$ can be 0 or 1; see the plots below.



In particular, both the topological multiplicity $\nu_h(X)$ and the topological degree $\deg(h, \text{BCJ}(\Gamma), X)$ are always zero, regardless of Γ ; this curious fact can be understood by considering X as a coalescence of p and $\bar{p}$ at which their topological multiplicities cancel.

Exercise 4.54. For the continuous map $h(z) = |z|^2 - i\bar{z}$:

- Find the roots of h .
- For each root a of h , choose a positively oriented Jordan curve Γ_a and compute the topological multiplicity $\nu_h(a)$.
- Let $\Gamma : [0, 1] \rightarrow \mathbb{C}$, $\Gamma(t) = 2e^{2\pi it}$. Find the image curve $h(\Gamma)$ and compute the topological degree of h at 0.
- Use (b,c) to verify (4.18). Gain a better understanding of this equality by observing that $h(z) = \bar{z}(z - i)$, and then mimicking the analysis of Example 4.35.